

EXAM 7 – SPRING 2011

10. (3.25 points)

Given the following information for a line of business for a reinsurance company as of December 31, 2010:

Calendar Accident Year	Earned Risk Pure Premium (\$000)	Adjusted Premium (\$000)	Aggregate Reported Loss (\$000)	Aggregate Loss Report Lag	Chain Ladder IBNR (\$000)
2005	9,200	10,000	9,500	95%	500
2006	10,100	12,000	8,000	85%	1,412
2007	11,500	11,500	7,300	72%	2,839
2008	12,300	13,000	6,500	55%	5,318
2009	13,500	14,100	4,800	40%	7,200
<u>2010</u>	<u>15,000</u>	<u>15,000</u>	<u>2,500</u>	<u>28%</u>	<u>6,429</u>
Total	71,600	75,600	38,600		23,698

a. (1.5 points)

Use the Stanard-Bühlmann method to estimate the total IBNR.

b. (0.5 point)

Briefly describe one advantage and one disadvantage of using the Stanard-Bühlmann method to calculate IBNR for a reinsurer.

c. (1.25 points)

Use a credibility-weighted combination of the Stanard-Bühlmann and chain ladder estimates with a linear function of the report lag as the credibility weight to calculate a total IBNR estimate. For the chain ladder estimate, use a credibility factor of 0.6.

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Sample Solutions
Exam 07, 2011, Question 10

Solution 1

a)

$$\begin{aligned}\text{SB ELR} &= \text{Reported Losses} / \Sigma (\text{Adjusted Premium}(k) \times \text{Report Lag}(k)) \\ &= 38,600 / (10,000 \times 0.95 + 12,000 \times 0.85 + 11,500 \times 0.72 + 13,000 \times 0.55 + 14,100 \times 0.40 + \\ &\quad 15,000 \times 0.28) \\ &= 0.858\end{aligned}$$

$$\begin{aligned}\text{SB IBNR} &= \text{SB ELR} \times \Sigma (\text{Adjusted Premium}(k) \times (1 - \text{Report Lag}(k))) \\ &= 0.858 \times (10,000 \times 0.05 + 12,000 \times 0.15 + 11,500 \times 0.28 + 13,000 \times 0.45 + 14,100 \times 0.60 + \\ &\quad 15,000 \times 0.72) \\ &= \mathbf{26,291}\end{aligned}$$

b)

SB method advantage: use actual claims data to derive ELR instead of judgmentally picking it like the BF.

SB method disadvantage: It's hard/difficult to obtain accurate adjusted on-level premium data.

c)

$$\begin{aligned}\text{Credibility Weighted IBNR} &= \Sigma \text{Credibility Factor} \times \text{Report Lag}(k) \times \text{CL IBNR}(k) + (1 - \\ &\quad \text{Credibility Factor}) \times \text{Report Lag}(k) \times \text{SB IBNR}(k) \\ &= (0.6 \times 0.95 \times 500 + (1 - 0.6 \times 0.95) \times 429) + (0.6 \times 0.85 \times 1412 + (1 - 0.6 \times 0.85) \times 1,545) + \\ &\quad (0.6 \times 0.72 \times 2839 + (1 - 0.6 \times 0.72) \times 2,764) + (0.6 \times 0.55 \times 5318 + (1 - 0.6 \times 0.55) \times 5,021) + \\ &\quad (0.6 \times 0.40 \times 7200 + (1 - 0.6 \times 0.40) \times 7,262) + (0.6 \times 0.28 \times 6429 + (1 - 0.6 \times 0.28) \times 9,270) \\ &= \mathbf{25,903}\end{aligned}$$